

Assignment 2: Related Work Review and Literature Analysis

Course: CSC-355 Natural Language Processing

Instructor: Dr. Muzamil Ahmed

Department: Computer Science, Namal University Mianwali

Session / Semester: 2022–2026 (8th)

Submission Date: 03-04-2026

Students:

- M. Raqib Hayat — NUM-BSCS-2022-40

- Abu Bakar — NUM-BSCS-2022-41

Project Title: *Robust Sentiment and Emotion Classification on Noisy Urdu Tweets:*

A Multi-Stage NLP Pipeline on SentiUrdu-1M.

1. Introduction and Scope of the Review

Sentiment and emotion analysis on Urdu social media text sits at the intersection of three difficult problems: a morphologically rich, right-to-left, under-resourced language; the noisy linguistic style of tweets (informal spelling, emoticons, code-mixing with English and Roman Urdu); and the absence of large, manually annotated corpora that has, until recently, forced researchers either to work on a few thousand carefully labelled sentences or to rely on weakly supervised labels assigned by emoticon heuristics and lexical resources such as SentiWordNet. The selected project tackles this triad directly, building a noise-aware multi-stage pipeline over the SentiUrdu-1M corpus and comparing classical machine learning, deep neural, and multilingual transformer families on a single fair evaluation set. To motivate that design, this review surveys twenty highly relevant studies drawn from IEEE, ACM, Springer, Elsevier, MDPI, Taylor & Francis, and reputable indexed conference venues, and organises them into three methodological generations that mirror the project’s own progression from feature engineering to deep architectures to large multilingual transformers.

Rather than summarising each paper in isolation, the discussion is grouped by the recurring questions that shape Urdu sentiment research: how to represent words in a script that lacks well-developed pre-trained embeddings, how to compensate for label scarcity through weak supervision while controlling for the resulting noise, and how to evaluate models fairly when the only large-scale labels available are themselves automatically generated. Earlier studies, almost exclusively classical and feature-engineered, prioritised lexicon coverage and shallow classifiers because they were the only models that could learn from datasets of a few hundred to a few

thousand documents. Later studies increasingly employed CNN and recurrent neural architectures as Urdu-specific word embeddings became available, and the most recent generation has shifted decisively to multilingual transformer models such as mBERT, XLM-R, and XLM-T, which offer strong contextual representations at the cost of demanding scale and engineering effort. Each generation has inherited the limitations of the previous one, and each leaves an open question that the proposed project must address: how to combine the scale of weakly supervised data with the robustness of clean evaluation, and how to compare model families on equal footing when reported accuracies in the literature span from below 60 % to over 96 % on different corpora.

2. Traditional Machine Learning Approaches

Early work on Urdu and Roman Urdu sentiment analysis was dominated by lexicon-based and classical supervised classifiers, motivated by two practical constraints: corpora were small enough to be labelled manually, and Urdu lacked the pre-trained embeddings and large GPU budgets that neural models require. Within this tradition, two recurring design choices structure most of the literature: whether to rely on a polarity lexicon such as a translated SentiWordNet, or on bag-of-words features fed to algorithms such as Naïve Bayes, Decision Trees, k-Nearest Neighbours, or Support Vector Machines. The earliest representative study of the supervised line, by Bilal, Israr, Shahid, and Khan, addressed the problem of automatically classifying Roman-Urdu blog opinions and proposed a straightforward pipeline in which 300 manually collected opinions (150 positive and 150 negative) drawn from blogs were represented as bag-of-words vectors and processed in WEKA using Naïve Bayes, Decision Tree (J48), and k-Nearest Neighbours classifiers [1]. The study reported that Naïve Bayes outperformed the other two with the highest accuracy, precision, recall, and F-measure on the held-out test set, a result that established Naïve Bayes as a default baseline in subsequent Roman-Urdu work. Its limitations, however, are characteristic of the entire first generation: a 300-sample dataset is too small to support claims about generalisation, the binary positive/negative scheme ignores neutral and mixed sentiment which dominate social media data, and the bag-of-words representation discards both word order and the heavy code-mixing typical of Pakistani online discourse.

A parallel strand of work approached the same problem at the sequence level rather than as document classification. Nasim and Ghani targeted Urdu Twitter rather than blog text and introduced a Markov-chain-based classifier that modelled sentiment as the probability of transitions among polarity states along a tweet’s word sequence [2]. The contribution of their study was twofold: a manually labelled three-class Urdu tweet corpus made publicly available on GitHub, and the demonstration that a sequence-aware probabilistic model outperformed both lexicon-based methods and standard supervised classifiers (SVM, Naïve Bayes) on the same data when evaluated with accuracy, precision, recall, and F1. Although the Markov-chain approach captures local dependencies that bag-of-words discards, its design assumes well-tokenised, lightly noised input and degrades quickly on the kind of emoji-laden, transliterated tweets that constitute most of SentiUrdu-1M; it also lacks any explicit mechanism for handling sarcasm or contextual negation.

Lexicon-based methods, which require no labelled data at the document level but depend critically on the coverage and polarity accuracy of an underlying word list, were systematically advanced by Mukhtar and Khan, who constructed a domain-balanced Urdu sentiment lexicon and combined it with grammar-aware rules for negation, intensification, and conjunction

handling [3]. Evaluated on a multi-domain Urdu corpus spanning movies, sports, books, and politics, their lexicon-based system outperformed both supervised classifiers trained on the same data and earlier lexicon baselines, with accuracy and F1 scores exceeding those of SVM and decision-tree models in three of four domains. The work makes a strong methodological case for hybrid lexicon-plus-rule systems in low-resource settings but inherits the central weakness of all lexicon methods: coverage is bounded by the size of the lexicon, novel out-of-vocabulary terms (which are continuously generated on social media) receive zero polarity, and emoticons, the dominant emotional cue in many tweets, are not modelled at all unless added by hand.

The breadth of the classical generation is captured in the survey by Khattak and colleagues, which is the most comprehensive review of Urdu sentiment analysis from an Elsevier-indexed venue and groups roughly five years of work into lexicon, machine learning, and hybrid families [4]. The survey identifies recurring evaluation metrics (accuracy, precision, recall, F1), recurring datasets (small manually annotated corpora of 1,000–10,000 items, plus the first emergence of weakly labelled data), and recurring limitations: the absence of standard benchmarks, inconsistent pre-processing, the rarity of three-class evaluation, and almost no work on emotion (multi-class) classification beyond positive/negative polarity. The survey is methodologically useful precisely because it foregrounds the gap that later transformer-based work would attempt to close, and it provides one of the few systematic baselines against which the present project’s results can be compared.

Two studies illustrate how the classical generation transitioned toward hybrid pipelines that anticipated the deep-learning shift. Muhammad and Burney, working with symmetric two-class Urdu datasets collected from Instagram, Twitter, and Facebook, applied a battery of machine-learning algorithms (Naïve Bayes, SVM, k-NN, Random Forest) alongside deep-learning baselines and a hybrid feature-extraction scheme that combined TF-IDF, n-grams, and selected linguistic features [5]. Their evaluation, reported with accuracy, precision, recall, and F1, established that hybrid feature pipelines could close part of the gap to deep models on small Urdu corpora but did not exceed them when data scaled. The study’s main limitation is its restriction to two-class symmetric data, which avoids the neutral class that dominates real Urdu tweet streams. Outside Urdu but methodologically influential, the ASTD dataset construction work by Nabil, Aly, and Atiya defined a four-way Twitter sentiment labelling scheme (objective, subjective-positive, subjective-negative, subjective-mixed) on roughly 10,000 Arabic tweets and provided classical-classifier baselines along with standard balanced and unbalanced splits [6]. ASTD is methodologically relevant because it codifies a labelling protocol and benchmark practice for a typologically related, similarly low-resource, right-to-left language, providing a template that subsequent Urdu corpus construction has implicitly followed.

Across this first generation, the recurring pattern is clear: classical pipelines achieve respectable accuracies (typically 70 – 85 % on small balanced corpora), are interpretable, and require modest computational resources, but their performance is bounded by lexicon coverage, by dataset size, and by their inability to model the order- and context-dependent phenomena (sarcasm, negation scope, code-mixing) that pervade Urdu social media.

3. Deep Learning Approaches

The second methodological generation responded directly to the representational limits of bag-of-words and lexicon systems by adopting dense word embeddings and neural sequence models.

Within Urdu and Roman Urdu, this generation is dominated by convolutional, recurrent (LSTM, GRU, BiLSTM), and increasingly hybrid CNN–RNN architectures, often combined with task-specific embeddings trained on local Urdu text. Ashraf, Khan, Butt, Chang, Sidorov, and Gelbukh provide a particularly important reference point because they shifted the task from binary or three-class polarity to multi-label emotion classification, releasing the first multi-label Urdu emotion dataset of 6,043 Nastaliq tweets annotated with six basic emotions and reporting results from a wide model sweep including Random Forest, Decision Tree (J48), Sequential Minimal Optimization, AdaBoost, Bagging, 1D-CNN, LSTM, LSTM with CNN features, and a BERT baseline [7]. Evaluation used the metrics appropriate to multi-label settings, namely micro and macro F1, Hamming loss, exact match, and accuracy, which exposed the inadequacy of accuracy alone for emotion problems where labels are highly imbalanced. The combination of CNN features feeding an LSTM, trained over stylometric and n-gram representations, was reported as the strongest classical-neural configuration, while the BERT baseline established a higher ceiling that subsequent transformer-focused work has continued to lift. The principal limitation of this study is the modest scale of the manually annotated dataset and the difficulty of generalising six-emotion taxonomies trained on a few thousand tweets to the open-domain content of corpora like SentiUrdu-1M.

For Roman Urdu, where the script difficulty is replaced by transliteration variance, Nagra and colleagues proposed a Faster Recurrent Convolutional Neural Network (Faster RCNN) variant trained on the RUSA-19 corpus and compared it against rule-based, n-gram, and conventional RCNN baselines for binary and three-class sentiment classification [8]. The proposed architecture combined convolutional feature extractors with recurrent layers and consistently exceeded the comparators in accuracy, precision, recall, and F1, particularly on the three-class setting where boundary tweets between negative and neutral classes had degraded the rule-based systems. The study is a useful demonstration that Roman-Urdu sentiment benefits from sequence-aware models, but it does not address transliteration normalisation, multilingual code-mixing, or label noise, and its results are bounded to a single corpus whose annotation reliability is not independently audited.

The cleanest demonstration of how deep architectures scale to Urdu beyond Twitter is provided by Khan, Amjad, Ashraf, and Chang, who manually annotated 9,312 Urdu user reviews drawn from food and beverages, movies and plays, software and apps, politics, and sports, and benchmarked rule-based, machine learning (SVM, Naïve Bayes, AdaBoost, MLP, Logistic Regression, Random Forest), and deep learning (1D-CNN, LSTM, BiLSTM, GRU, BiGRU) classifiers across word n-grams, character n-grams, pre-trained FastText vectors, and BERT word embeddings [9]. Their fine-tuned multilingual BERT model achieved an F1 of 81.49 %, outperforming both classical and deep baselines and establishing one of the strongest open three-class Urdu sentiment benchmarks. The principal limitation of the study, acknowledged by the authors, is that performance is reported on a curated review corpus rather than on noisy tweets, leaving open the question of how the same architectures behave under the emoticon-heavy, code-mixed conditions of SentiUrdu-1M.

Hybrid linguistic–neural approaches have continued to gain ground because they offer a route to robustness without abandoning the explanatory power of deep models. Sehar and colleagues developed a concept-level sentiment framework for Urdu YouTube reviews that combined dependency-grammar–based rules with a deep neural classifier, using the rule system to trigger sentiment flow between syntactically related words and falling back to the DNN whenever no

rule fired [10]. Evaluated on a YouTube-sourced Urdu dataset spanning movies, politics, and commercial products, the framework was reported to surpass LSTM, CNN, SVM, Logistic Regression, and MLP baselines by 6–7 percentage points in accuracy and F1. The study’s main limitation is that the hand-crafted dependency rules are language- and domain-specific, scaling poorly to the open vocabulary of Twitter, and the dataset itself is comparatively small relative to weakly labelled tweet corpora. A more recent attention-based variant by Khan, Qazi, Chang, Alhajlah, and Mahmood proposed an attention-augmented stacked two-layer CNN–BiLSTM DNN with multilingual BERT embeddings as input, achieving 83.12 % and 78.91 % accuracy on the UCSA-21 and UCSA Urdu sentiment datasets respectively [11]. The architecture demonstrates that stacking BiLSTMs and adding an attention layer over BERT-derived features captures sequential dependencies more effectively than either component alone, but the study does not analyse robustness to noisy labels or out-of-domain tweet data, leaving an evaluation gap that the present project intends to fill.

Two further studies extend the deep-learning generation along complementary directions. Samreen and Ali contributed a Roman-Urdu lexicon and annotation resource of more than 14,000 emotion entries covering nine emotions with polarity and mood, transliterated from the NRC Urdu emotion lexicon and verified against three independent Urdu dictionaries [12]. They automatically annotated approximately 10,000 social-media sentences using unigram-feature matching and string-distance estimation against the lexicon. Because this is a data article rather than a modelling paper, the relevant evaluation measures are dataset quality indicators rather than classifier metrics: transliteration accuracy verified against three reference dictionaries, lexicon coverage across the nine emotion categories, and the dataset’s class-distribution statistics. The resource enables emotion-aware learning on Roman Urdu and exemplifies the kind of automatic annotation pipeline that the present project will adapt for tweet-level labelling; its main limitation is that the automatic labels inherit the systematic bias of the underlying lexicon, with no inter-annotator agreement reported for the resulting sentence-level annotations. At the multilingual end of the deep-learning literature, Barreto, Moura, Carvalho, Paes, and Plastino conducted a systematic assessment of sentiment classification on twenty-two tweet datasets across multiple languages, comparing classical bag-of-words and TF-IDF representations against static embeddings and contextualised representations from Transformer-based models including BERT and BERTweet [13]. Their evaluation, reported with accuracy and macro F1, demonstrated that fine-tuned contextual encoders consistently outperform classical representations on noisy tweet data but that the magnitude of the gain depends strongly on dataset size and domain match. The work supplies a methodological template for the comparative evaluation that the present project will conduct on SentiUrdu-1M.

Taken together, deep-learning approaches improved accuracy by ten to twenty points over classical baselines on comparable Urdu corpora and unlocked emotion and aspect tasks that lexicon and bag-of-words methods could not address, but they continue to rely on relatively small, manually labelled datasets and their reported accuracies remain optimistic when transferred to weakly labelled, emoji-heavy tweet streams.

4. Transformer-Based and Recent Advanced Methods

The third methodological generation departs from sequence-specific architectures and instead fine-tunes large multilingual transformer models, principally multilingual BERT (mBERT) and XLM-RoBERTa (XLM-R), on Urdu-specific data, occasionally augmented with domain-adapted

pre-training on multilingual tweet corpora. The flagship resource that anchors this generation, and the dataset around which the present project is built, is SentiUrdu-1M, contributed by Ghafoor, Imran, Daudpota, Kastrati, Shaikh, and Batra and published in PLOS ONE [14]. The study assembles 1,140,821 Urdu tweets, the first publicly available Urdu tweet corpus of this scale, and addresses the labelling bottleneck through a weakly supervised pipeline that combines emoticon polarity heuristics with SentiWordNet-derived word-level scores to assign positive, negative, and neutral labels automatically. Baseline experiments fine-tune deep learning models on the resulting labels and compare three labelling strategies (VADER, TextBlob, and the proposed emoticon-plus-SentiWordNet approach), showing that VADER and TextBlob overproduce neutral labels because they ignore emoticons whereas the proposed scheme yields a substantially more discriminative label distribution. The study's central methodological contribution is its honest treatment of label noise as a first-class concern. It stops short, however, of investigating noise-robust training objectives, label-cleaning algorithms, or evaluation on a clean manually verified held-out set. Those three omissions together define the central research gap that the proposed project will address.

Building directly on SentiUrdu-1M, Narejo and colleagues investigated whether explicit emoji modelling, rather than implicit emoticon use in labelling, could improve transformer-based classification on the same corpus [15]. Their pipeline fine-tunes mBERT and XLM-RoBERTa on text-only tweets and reports baseline accuracies of 64 % and 65 % respectively, which is consistent with the difficulty of three-class classification under weakly supervised labels at the one-million-tweet scale. The study then augments the text encoder with a 1,194-emoji vocabulary represented through pre-trained emoji2vec and a proposed Urdu-specific FastText emoji embedding, and explores three fusion strategies (concatenation, neural network fusion, and attention-based fusion). Neural fusion with the Urdu-specific FastText emoji embeddings raises accuracy to as high as 95 % for the best configuration. The result is striking but should be interpreted with care: because the labels of SentiUrdu-1M are themselves derived in part from emoticons, training models that read emojis effectively learns the labelling rule, and the headline accuracy may overstate generalisation to tweets without emoji cues. The study nevertheless demonstrates the value of multimodal input fusion in Urdu sentiment analysis and motivates careful ablation studies that the present project will replicate on a clean evaluation subset.

Beyond polarity, transformer-based hybrids are now driving progress on the most context-sensitive Urdu tasks. Hassan, Hussain, Maab, Habib, Khan, and Masood addressed Urdu sarcasm detection, an inherently context-dependent problem that defeats lexicon and bag-of-words approaches, and constructed a manually re-annotated dataset of 12,910 tweets derived from a 19,995-tweet public Urdu corpus [16]. After establishing baselines using CNN, LSTM, GRU, BiLSTM, and CNN-LSTM trained on fastText embeddings, they proposed a hybrid that injects mBERT contextual embeddings into a BiLSTM enriched with multi-head attention. The mBERT-BiLSTM-MHA model achieved an accuracy of 79.51 % and an F1 of 80.04 %, outperforming every pure deep-learning baseline. The work illustrates how transformer encoders, attention, and sequence models can be composed for context-sensitive tasks. Its principal limitation, common to all sarcasm studies, is that the model's reliability depends on the quality of the re-annotation and on the relatively narrow tweet selection from which the corpus was drawn.

Pure-transformer fine-tuning has also been pushed to high accuracies on cleaner Urdu data. Saeed, Ahmed, Ali, Ramzan, Mohib, Bagga, Rahman, and Khan curated and annotated Urdu

blog content into positive, neutral, and negative classes, evaluated five machine-learning classifiers (SVM, k-NN, Naïve Bayes, Logistic Regression, Multinomial Logistic Regression), and fine-tuned mBERT on the resulting dataset [17]. The fine-tuned mBERT achieved an accuracy of 96.5 % on the test set, substantially outperforming all classical classifiers and confirming that contextual representations pre-trained on 104 languages transfer effectively to Urdu when the labels are clean. The headline accuracy must, however, be read against the domain gap: blog text is well-formed and stylistically distinct from tweets, and the same model fine-tuned on noisy SentiUrdu-1M labels typically reports accuracies in the mid-60s rather than the mid-90s. That gap motivates the project’s emphasis on a clean held-out evaluation set distinct from the weakly labelled training corpus.

A complementary direction within this generation explores weak supervision and transfer learning rather than larger fine-tuning sets. Zoya proposed a weakly supervised aspect-based sentiment analysis framework for Urdu tweets that requires only a small set of seed words per aspect and sentiment class and trains a joint aspect–sentiment topic-embedding model coupled with a CNN/BiLSTM classifier through iterative self-training on unlabelled data [18]. The framework was evaluated using F1 on a held-out portion of the unlabelled tweet corpus and demonstrated that competitive ABSA performance is achievable without manually labelled aspect annotations, which is directly relevant to the project’s interest in noise-robust learning. Limitations include sensitivity to seed-word choice and the cumulative effect of self-training errors, both of which require careful audit. At a larger scale, Barbieri, Espinosa Anke, and Camacho-Collados introduced XLM-T, an XLM-R variant continually pre-trained on millions of multilingual tweets across more than thirty languages, together with a unified set of sentiment-analysis datasets in eight languages [19]. Their evaluations, reported in accuracy and macro F1, established XLM-T as a strong cross-lingual baseline that consistently outperforms generic XLM-R on Twitter sentiment tasks because its pre-training corpus matches the target domain. Although Urdu is not among the eight evaluation languages, XLM-T’s design implies that Urdu sentiment models initialised from XLM-T should outperform mBERT and XLM-R on tweet data, a hypothesis the present project will test directly.

Recent multimodal and multilingual work is beginning to extend transformer-based sentiment analysis beyond text. Thakkar, Hakimov, and Tadić proposed M2SA, a multimodal multilingual sentiment-analysis pipeline for Twitter that fuses text encoders (large language models) with image encoders and evaluates on Twitter datasets in twenty-one languages [20]. The pipeline was evaluated using accuracy and macro F1 across the per-language test sets, with unimodal text-only and combined text-plus-image configurations compared head-to-head. The study found that a sentiment-tuned LLM text encoder is exceptionally effective on its own and that adding image features yields measurable gains primarily when the text is short or ambiguous. The principal limitation is that genuine multimodal supervision is expensive to assemble for every target language, and Urdu coverage in the released benchmark is partial. Although the project does not currently plan to use image data, M2SA’s design demonstrates how additional modalities, including emojis treated as a quasi-visual channel, can be integrated into a transformer pipeline without sacrificing text-encoder quality.

Several convergent findings emerge from the transformer-based literature. Fine-tuned mBERT and XLM-R routinely outperform classical and pure-RNN baselines on Urdu sentiment, with reported gains of 5–25 points depending on dataset cleanliness. Headline accuracies are also extremely sensitive to label quality: 96 % on clean blog data and 64 % on weakly labelled tweets

are not contradictions but evidence that the same architecture is bounded by its training labels. Most strikingly, almost no published work on Urdu has paired large weakly supervised training data with a clean manually verified held-out test set, which is precisely the experimental design the present project proposes.

5. Comparative Summary Table

Sr. No	Ref	Year	Method	Dataset	Findings / Accuracy	Limitations
1	[1]	2016	Naïve Bayes, Decision Tree (J48), k-NN with BoW features in WEKA	300 Roman-Urdu / English blog opinions (150 +ve, 150 -ve)	Naïve Bayes best on accuracy, precision, recall, F-measure	Tiny corpus; binary only; no neutral class; no code-mixing handling
2	[2]	2020	Markov-chain sequence model vs. lexicon / classical ML	Manually labelled Urdu tweet corpus (three-class)	Outperforms lexicon-based and classical ML baselines on accuracy / F1	Sensitive to noise and tokenisation; no sarcasm or negation modelling
3	[3]	2020	Lexicon + grammar rules (negation, intensifier, conjunction)	Multi-domain Urdu corpus (movies, sports, books, politics)	Lexicon-based outperforms supervised ML in three of four domains	Bounded by lexicon coverage; ignores emoticons and OOV tokens
4	[4]	2021	Survey of lexicon, ML, hybrid methods	N/A (review paper)	Catalogues recurring datasets, metrics, limitations in Urdu SA	No new experiments; gaps in standard benchmarking acknowledged
5	[5]	2023	TF-IDF / n-gram features with SVM, NB, k-NN, RF + DL baselines	Symmetric two-class Urdu social-media data (IG, Twitter, FB)	Hybrid features narrow gap to DL on small symmetric data	Two-class only; no neutral; no large-scale evaluation
6	[6]	2015	Classical ML baselines on Twitter sentiment	ASTD (~10,000 Arabic tweets, 4-way labels)	Establishes benchmark for 4-way Arabic Twitter sentiment	Arabic only; classical baselines; small relative to modern corpora
7	[7]	2022	RF, J48, SMO,	6,043 Urdu	Micro/macro	Modest scale;

			AdaBoost, Bagging, 1D-CNN, LSTM, LSTM+CNN, BERT (multi-label)	Nastaliq tweets, six emotions	F1, Hamming loss, exact match reported; LSTM+CNN strong; BERT competitive	multi-label evaluation only; limited transfer to noisy tweet streams
8	[8]	2022	Faster Recurrent Convolutional Neural Network (Faster RCNN)	RUSA-19 Roman-Urdu corpus	Outperforms rule-based, n-gram, RCNN for binary and three-class	Single corpus; no transliteration normalisation; no noise analysis
9	[9]	2022	Rule-based, SVM/NB/AdaBoost/MLP/LR/RF, 1D-CNN/LSTM/BiLSTM/GRU/Bi-GRU, fine-tuned mBERT	9,312 manually annotated Urdu reviews (food, movies, apps, politics, sports)	mBERT F1 = 81.49 %, exceeds ML and DL baselines	Curated review text, not noisy tweets; manual labels do not scale
10	[10]	2023	Dependency-rule trigger + DNN fallback (concept-level SA)	Urdu YouTube reviews on movies, politics, products	Surpasses LSTM, CNN, SVM, LR, MLP by 6–7 % accuracy / F1	Rules are language- and domain-specific; small dataset
11	[11]	2024	Attention-based stacked two-layer CNN-BiLSTM with mBERT embeddings	UCSA-21 and UCSA Urdu sentiment datasets	83.12 % and 78.91 % accuracy respectively	No noise-robustness analysis; not evaluated on tweets
12	[12]	2024	Lexicon transliteration (NRC Urdu → Roman Urdu) + unigram/distance auto-annotation	14,000+ Roman-Urdu emotion lexicons; ~10,000 auto-annotated sentences	Provides nine-emotion lexicon with polarity and mood	Data article (no modelling); auto-annotation introduces noise
13	[13]	2021	BoW, TF-IDF, static and contextual embeddings, BERT/BERTweet	22 multilingual tweet datasets	Contextual encoders consistently best on noisy tweets; magnitude depends on data	Urdu not directly included; size- and domain-dependent gains

14	[14]	2023	Weakly supervised labelling (emojis + SentiWordNet) + DL baselines	SentiUrdu-1M (1,140,821 Urdu tweets, three-class)	First large-scale Urdu tweet corpus; emoji scheme beats VADER/TextBlob neutral bias	No noise-robust training; no clean held-out evaluation set
15	[15]	2024	mBERT / XLM-R fine-tuning + emoji2vec / Urdu-FastText emoji fusion	SentiUrdu-1M + 1,194 emojis	Text-only ~64–65 %; NN-fusion with Urdu emoji embeddings up to ~95 %	Labels partly derived from emojis; risk of inflated headline accuracy
16	[16]	2024	mBERT-BiLSTM-MHA hybrid for sarcasm	12,910 manually re-annotated tweets from 19,995-tweet Urdu corpus	Acc = 79.51 %, F1 = 80.04 %, beats CNN/LSTM/GRU/BiLSTM/CNN-LSTM	Single re-annotated source; narrow domain; no multi-class emotion
17	[17]	2024	SVM, k-NN, NB, LR, MLR, fine-tuned mBERT	Manually annotated Urdu blog data (three-class)	mBERT accuracy = 96.5 %, far above classical baselines	Clean blog domain; will not generalise directly to noisy tweets
18	[18]	2023	Weakly supervised joint aspect-sentiment topic embedding + CNN/BiLSTM self-training	Unlabelled Urdu tweet corpus + small seed-word lists	Competitive ABSA F1 without manual aspect labels	Sensitive to seed words; self-training error propagation
19	[19]	2022	XLM-R continually pre-trained on multilingual tweets (XLM-T)	30+ languages of tweets; unified eight-language sentiment benchmark	XLM-T beats generic XLM-R on Twitter sentiment; strong zero-shot cross-lingual baseline	Urdu not in eight-language evaluation set; needs adaptation
20	[20]	2024	Multimodal LLM text encoder + image encoder fusion (M2SA)	21-language Twitter sentiment datasets, text + image	Sentiment-tuned LLM strongest text encoder; image features help when text is ambiguous	Multimodal data requirement; Urdu coverage partial

6. Identified Research Gap

The literature surveyed exhibits a consistent and exploitable gap. Large-scale resources for Urdu tweet sentiment now exist, most notably SentiUrdu-1M [14] and the emoji-enhanced variant in [15]. Their labels, however, are weakly supervised, derived from emoticon heuristics and SentiWordNet matching, and therefore carry systematic, learnable noise that standard supervised pipelines treat as ground truth. In parallel, the small, manually annotated Urdu corpora used by the strongest deep-learning and transformer studies ([7], [9], [10], [11], [16], [17]) produce headline accuracies in the 80 – 96 % range that do not transfer to large noisy tweet streams, where the same architectures typically report accuracies in the mid-60s ([15]). Almost no published Urdu study (i) trains a transformer model on a million-scale weakly labelled corpus while explicitly modelling label noise, (ii) evaluates the resulting model on a small, independently verified clean held-out test set drawn from the same domain, and (iii) compares classical, deep, and transformer families on equal footing under both noisy and clean evaluation. Furthermore, the integration of Urdu emoji semantics, code-mixing with English and Roman Urdu, and contextual sarcasm (each addressed in isolation by [15], [12], and [16] respectively) has not been examined within a single noise-aware framework. The proposed project addresses these gaps by constructing a stratified subset of SentiUrdu-1M, training TF-IDF + Logistic Regression / SVM, CNN / BiLSTM, and fine-tuned mBERT / XLM-R models within the same pipeline, evaluating on a clean held-out subset of 500–1,000 manually verified tweets, and analysing the gap between weakly supervised and clean-label accuracy as a direct quantitative measure of label noise impact.

7. References (IEEE Format)

- [1] M. Bilal, H. Israr, M. Shahid, and A. Khan, “Sentiment classification of Roman-Urdu opinions using Naïve Bayesian, Decision Tree and KNN classification techniques,” *Journal of King Saud University – Computer and Information Sciences*, vol. 28, no. 3, pp. 330–344, 2016.
- [2] Z. Nasim and S. Ghani, “Sentiment analysis on Urdu tweets using Markov chains,” *SN Computer Science*, vol. 1, no. 5, art. no. 269, 2020.
- [3] N. Mukhtar and M. A. Khan, “Effective lexicon-based approach for Urdu sentiment analysis,” *Artificial Intelligence Review*, vol. 53, no. 4, pp. 2521–2548, 2020.
- [4] A. Khattak, M. Z. Asghar, A. Saeed, I. A. Hameed, S. A. Hassan, and S. Ahmad, “A survey on sentiment analysis in Urdu: A resource-poor language,” *Egyptian Informatics Journal*, vol. 22, no. 1, pp. 53–74, 2021.
- [5] K. B. Muhammad and S. M. A. Burney, “Innovations in Urdu sentiment analysis using machine and deep learning techniques for two-class classification of symmetric datasets,” *Symmetry*, vol. 15, no. 5, art. no. 1027, 2023.
- [6] M. Nabil, M. Aly, and A. F. Atiya, “ASTD: Arabic sentiment tweets dataset,” in *Proc. 2015 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, Lisbon, Portugal, 2015, pp. 2515–2519.
- [7] N. Ashraf, L. Khan, S. Butt, H.-T. Chang, G. Sidorov, and A. Gelbukh, “Multi-label emotion classification of Urdu tweets,” *PeerJ Computer Science*, vol. 8, art. no. e896, 2022.

- [8] A. A. Nagra, K. Alissa, T. M. Ghazal, S. Kukunuru, M. M. Asif, and M. Fawad, "Deep sentiments analysis for Roman Urdu dataset using faster recurrent convolutional neural network model," *Applied Artificial Intelligence*, vol. 36, no. 1, art. no. 2123094, 2022.
- [9] L. Khan, A. Amjad, N. Ashraf, and H.-T. Chang, "Multi-class sentiment analysis of Urdu text using multilingual BERT," *Scientific Reports*, vol. 12, art. no. 5436, 2022.
- [10] U. Sehar, S. Kanwal, N. I. Allheeib, S. Almari, F. Khan, K. Dashtipur, M. Gogate, and O. A. Khashan, "A hybrid dependency-based approach for Urdu sentiment analysis," *Scientific Reports*, vol. 13, art. no. 22075, 2023.
- [11] L. Khan, A. Qazi, H.-T. Chang, M. Alhajlah, and A. Mahmood, "Empowering Urdu sentiment analysis: an attention-based stacked CNN-Bi-LSTM DNN with multilingual BERT," *Complex & Intelligent Systems*, vol. 11, art. no. 10, 2025.
- [12] A. Samreen and S. A. Ali, "Dataset construction to detect human behavior with the help of emotions, sentiments and mood for Roman Urdu," *Data in Brief*, vol. 52, art. no. 109906, 2024.
- [13] S. Barreto, R. Moura, J. Carvalho, A. Paes, and A. Plastino, "Sentiment analysis in tweets: An assessment study from classical to modern text representation models," *Language Resources and Evaluation*, vol. 57, no. 4, pp. 1727–1758, 2023.
- [14] A. Ghafoor, A. S. Imran, S. M. Daudpota, Z. Kastrati, S. Shaikh, and R. Batra, "SentiUrdu-1M: A large-scale tweet dataset for Urdu text sentiment analysis using weakly supervised learning," *PLOS ONE*, vol. 18, no. 8, art. no. e0290779, 2023.
- [15] K. R. Narejo, H. Zan, D. Oralbekova, K. P. Dharmani, M. Orken, and K. Mukhsina, "Enhancing emoji-based sentiment classification in Urdu tweets: Fusion strategies with multilingual BERT and emoji embeddings," *IEEE Access*, vol. 12, pp. 126587–126603, 2024.
- [16] M. E. Hassan, M. Hussain, I. Maab, U. Habib, M. A. Khan, and A. Masood, "Detection of sarcasm in Urdu tweets using deep learning and transformer based hybrid approaches," *IEEE Access*, vol. 12, pp. 61542–61555, 2024.
- [17] M. Saeed, N. Ahmed, D. Ali, M. Ramzan, M. Mohib, K. Bagga, A. U. Rahman, and I. M. Khan, "In-depth Urdu sentiment analysis through multilingual BERT and supervised learning approaches," *ICCK Transactions on Intelligent Systematics*, vol. 1, no. 3, pp. 161–175, 2024.
- [18] Zoya, "Weakly supervised learning for aspect-based sentiment analysis of Urdu tweets," in *Proc. Student Research Workshop, Recent Advances in Natural Language Processing (RANLP)*, 2023, pp. 70–78.
- [19] F. Barbieri, L. Espinosa Anke, and J. Camacho-Collados, "XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond," in *Proc. 13th Conf. on Language Resources and Evaluation (LREC)*, Marseille, France, 2022, pp. 258–266.
- [20] G. Thakkar, S. Hakimov, and M. Tadić, "M2SA: Multimodal and multilingual model for sentiment analysis of tweets," in *Proc. 2024 Joint Int. Conf. on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, Torino, Italy, 2024.